1 Convergence of Fourier Series

1.1 Convolution

Theorem 1.1.1 (Properties of the Convolution):

(1) For $f, g \in C^0(S^1)$, $f * g \in C^0(S^1)$

(2) Convolution is linear in each variable.

(3) For $f, g \in C^0(S^1)$, $f * g = g * f$

(4) For $f, g, h \in C^0(S^1)$, $(f * g) * h = f * (g * h)$

(5) For $f \in C^1(S^1), g \in C^0(S^1)$, $f * g \in C^1(S^1)$ and:

$$\frac{\mathrm{d}}{\mathrm{d}x}(f * g) = \left(\frac{\mathrm{d}f}{\mathrm{d}x} * g\right)$$

(6) For $f, g \in C^0(S^1)$, $\widehat{f * g}(n) = \hat{f}(n)\hat{g}(n)$

Proof,

(1) Since, $f, g \in C^0(S^1) \implies f(x-t)g(t)$ is a continuous function of $t \in \mathbb{R}$.

Moreover,

$$f(x-t+1)g(t+1) = f(x-t)g(t)$$

Hence, $f(x-t)g(t) \in C^0(S^1)$.

Consider,

$$f * g(x) = \int_0^1 \mathrm{d}t\, f(x-t)g(t)$$

which is defined because of continuity of the integrand.

Let $\varepsilon \in \mathbb{R}^+$.

Suppose, $x, y \in \mathbb{R} : |x-y| < \delta$.

So,

$$|f * g(x) - f * g(y)| = \left| \int_0^1 \mathrm{d}t\, f(x-t)g(t) - \int_0^1 \mathrm{d}t\, f(y-t)g(t) \right|$$
$$\leq \int_0^1 \mathrm{d}t\, |f(x-t) - f(y-t)||g(t)|$$

By continuity of g and compactness of S^1, it follows that $\sup_{x \in S^1} |g(x)| = M$ exists. Similarly, continuity of f and it being in S^1 means f is uniformly continuous on $\mathbb{R}$. Thus, we can chose δ such that:

$$\forall x, y \in \mathbb{R}, |x-y| < \delta \implies |f(x) - f(y)| < \frac{\varepsilon}{M}$$

So,

$$|f * g(x) - f * g(y)| < \left(\frac{\varepsilon}{M}\right)(M) = \varepsilon$$

and so we're done.

(2) Linearity of convolution is a trivial result which follows from the linearity of the integral.

(3) Now, here $f, g \in C^0(S^1)$, we have to show that the convolution is commutative.

$$f * g(x) = \int_0^1 \mathrm{d}t\, f(x-t)g(t)$$

Using the change of variable, $u = x - t$.

$$\mathrm{d}u = -\,\mathrm{d}t \quad t = x - u \quad u(0) = x \quad u(1) = x - 1$$

Hence, we get:

$$f * g(x) = \int_x^{x-1} \mathrm{d}t\, -f(u)g(x-u) = \int_{x-1}^{x} \mathrm{d}t\, g(x-u)f(u)$$

By periodicity, we can transform the integral into:

$$\int_0^1 \mathrm{d}u\, g(x-u)f(u) = g * f(x)$$

Therefore,

$$f * g(x) = g * f(x)$$

The rest also fall similarly. With (5) requiring an application of the MVT.

□

1.2 Fourier Inversion Theorem

Theorem 1.2.1: If (K_N) be a Dirac kernel and $f \in C^0(S^1)$, then:

$$\lim_{N\to\infty} (f * K_N)(x) = f(x)$$

where convergence is uniform on S^1.

Proof,

To prove this theorem, the key trick is the concentration property of the Dirac kernel which states that for any $\delta \in \mathbb{R}^+, \delta < \frac{1}{2}$,

$$\lim_{N\to\infty} \int_{\delta\leq|x|\leq\frac{1}{2}} \mathrm{d}x\, K_N(x) = 0$$

To apply this integral, we first break the convolution:

$$f * K_N(x) = \int_{|t|\leq\delta} \mathrm{d}t\, f(x-t)K_N(t) + \int_{\delta\leq|t|\leq\frac{1}{2}} \mathrm{d}t\, f(x-t)K_N(t)$$

Consider the integral:

$$\left| \int_{\delta\leq|t|\leq\frac{1}{2}} \mathrm{d}t\, f(x-t)K_N(t) \right| \leq \sup_{\delta\leq|t|\leq\frac{1}{2}} |f(x-t)| \int_{\delta\leq|t|\leq\frac{1}{2}} \mathrm{d}t\, K_N(t)$$

where we used the boundedness of f to pull out the sup and the positivity of K_N to remove the absolute value.

By the concentration property of the kernel, the limit of the RHS as $N \to \infty$ is 0. Hence, by squeezing,

$$\lim_{N\to\infty} \left| \int_{\delta \leq |t| \leq \frac{1}{2}} \mathrm{d}t \, f(x-t) K_N(t) \right| = 0$$

As we wish for uniform convergence, we can't make any assumption on x. The work we have done above has allowed us to estimate one of the two sums in the convolution. Now, we need to deal with the:

$$\int_{|t|\leq\delta} \mathrm{d}t \, f(x-t) K_N(t)$$

integral.

I want to claim that:

$$\lim_{N\to\infty} \int_{|t|\leq\delta} \mathrm{d}t \, f(x-t) K_N(t) = f(x)$$

This is really easy if we could use:

$$\int_{-\delta}^{\delta} \mathrm{d}t \, (f(x-t) - f(x)) K_N(t)$$

Since, by uniform continuity, we could make the difference as small we we want it to be. Say ε. So, we get:

$$\left| \int_{-\delta}^{\delta} \mathrm{d}t \, (f(x-t) - f(x)) K_N(t) \right| < \varepsilon \left(\int_{-\delta}^{\delta} \mathrm{d}t \, K_N(t) \right) < \varepsilon(1) = \varepsilon$$

Fortunately, we recall that

$$\lim_{N\to\infty} \left| \int_{\delta \leq |t| \leq \frac{1}{2}} \mathrm{d}t \, f(x-t) K_N(t) \right| = 0$$

only required the boundedness of $f(x-t)$. Since, $f(x-t)-f(x)$ is also bounded, it follows that:

$$\lim_{N\to\infty}\left|\int_{\delta\le|t|\le\frac{1}{2}} \mathrm{d}t\,(f(x-t)-f(x))K_N(t)\right| = 0$$

Therefore,

$$|f*K_N(x)-f(x)| = \left|\int_0^1 \mathrm{d}t\, f(x-t)K_N(t) - \int_0^1 \mathrm{d}t\, f(x)K_N(t)\right|$$

where, the integral follows by the unit mass property of K_N.

Splitting the integral, we get the inequality:

$$\begin{aligned}|f*K_N(x)-f(x)| \le &\left|\int_{-\delta}^{\delta} \mathrm{d}t\,(f(x-t)-f(x))K_N(t)\right| \\ &+\left|\int_{\delta\le|t|\le\frac{1}{2}} \mathrm{d}t\,(f(x-t)-f(x))K_N(t)\right|\end{aligned}$$

Since we can make δ as small as we wish, we choose δ such that:

$$|t|\le\delta \implies |f(x-t)-f(x)|<\varepsilon$$

where the above inequality follows by uniform continuity of f which follows by continuity and compactness.

So,

$$|f*K_N(x)-f(x)| \le \varepsilon + \left|\int_{\delta\le|t|\le\frac{1}{2}} \mathrm{d}t\,(f(x-t)-f(x))K_N(t)\right|$$

We now take the sup over x, by boundedness it is valid and it gives us:

$$\sup_x|f*K_N(x)-f(x)| \le \varepsilon + \sup_x\left|\int_{\delta\le|t|\le\frac{1}{2}} \mathrm{d}t\,(f(x-t)-f(x))K_N(t)\right|$$

Since, $\lim_{N\to\infty}\sup_x\left|\int_{\delta\leq|t|\leq\frac{1}{2}}\mathrm{d}t\,(f(x-t)-f(x))K_N(t)\right|=0$ because the limit didn't depend on x. More accurately,

$$\sup_x\left|\int_{\delta\leq|t|\leq\frac{1}{2}}\mathrm{d}t\,(f(x-t)-f(x))K_N(t)\right|\leq 2\sup_x|f(x)|\int_{\delta\leq|t|\leq\frac{1}{2}}\mathrm{d}t\,K_N(t)$$

Thanks to this, if we take the limsup over N, we get:

$$\limsup_{N\to\infty}\sup_x|f*K_N(x)-f(x)|\leq\varepsilon$$

Since, ε was arbitrary, it follows that:

$$f*K_N\rightrightarrows f$$

□

Corollary 1.2.1.1: For $f\in C^0(S^1)$, let $s_{N(x)}$ be the Nth Cesaro sum:

$$s_N(x)=(f*F_N)(x)=\frac{1}{N}\sum_{n=0}^{N-1}f_{n(x)}$$

of the Fourier series of f. Then,

$$s_n\rightrightarrows f\text{ on }S^1$$

Proof,

Since, the Fejer kernel is a Dirac kernel. It follows that,

$$(f*F_N)\rightrightarrows f\text{ on }S^1$$

so we're done.

□

Theorem 1.2.2 (Inversion Theorem of Fourier Series): For $f \in L^2(S^1)$, if f_N is the Nth Fourier Polynomial of f, then:

$$\lim_{N\to\infty} \|f - f_N\|_{L^2} = 0$$

Proof,

Let $\varepsilon \in \mathbb{R}^+$.

Since, $C^0(S^1)$ is dense in $L^2(S^1)$, it follows that there exists a function g such that:

$$\|f - g\| < \frac{\varepsilon}{2}$$

Now, define:

$$S_M = g * F_M$$

where, F_M is the Mth Fejer Kernel.

So, by triangle inequality.

$$\|f - S_M\| \leq \|f - g\| + \|g - S_M\|$$

Furthermore, since, S_M is technically an Mth order trigonometric polynomial, it follows by the best approximation lemma that:

$$\|f - f_M\| \leq \|f - S_M\| \leq \|f - g\| + \|g - S_M\|$$

Since, $S_M \rightrightarrows g \implies \|g - S_M\| \to 0$. Therefore, it follows that there exists a $M \in \mathbb{N}$ such that,

$$\|g - S_M\| < \frac{\varepsilon}{2}$$

Hence, $N \geq M$, we get, again by Best Approximation, that:

$$\|f - f_N\| \leq \|f - f_M\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Therefore, it follows that:

$$\lim_{N\to\infty} \|f - f_N\| = 0$$

□

Remark: The inversion theorem *only* guarantees L^2 convergence. It doesn't mention anything about uniform convergence nor even pointwise convergence. They need additional assumptions.

□

Now, we have Corollaries of the inversion theorem.

Corollary 1.2.2.1 *(Consequences of Inversion)*: Let $f \in L^2(S^1)$, then we have the following:

1. $(e_n)_{n\in\mathbb{Z}}$ forms an orthogonal basis.
2. (Parseval's Theorem) If $f, g \in L^2(S^1)$, we've:
$$\langle f, g\rangle = \sum_{n\in\mathbb{Z}} \hat{f}(n)\overline{\hat{g}(n)}$$
3. (Parseval's Theorem) If $f \in L^2(S^1)$, we've:
$$\|f\|^2 = \sum_{n\in\mathbb{Z}} \left|\hat{f}(n)\right|^2$$
4. $f \in L^2(S^1)$, if $\hat{f}(n) = 0 \forall n \in \mathbb{N} \implies f = 0$ a.e.

Proof,

Isomorphism theorem for Fourier series.

□

Corollary 1.2.2.2: For $f, g \in L^2(S^1)$, the following are equivalent:

1. $f = g$ a.e on S^1
2. $\hat{f}(n) = \hat{g}(n)$ for all $n \in \mathbb{Z}$

Proof,

Here, if $f = g$ ae this implies that $f - g = 0$ ae on $L^2(S^1)$. Therefore,

$$\widehat{f-g}(n) = 0 \implies \hat{f} = \hat{g}$$

for all $n \in \mathbb{Z}$.

The other direction is difficult. However, it's a corrolary of (4) of Corollary 1.2.2.1.

□

Lemma 1.2.3 (Extra Derivative Lemma): If $g \in L^2(S^1)$, then the two-sided series:

$$\sum_{n\neq 0}\left(\frac{1}{2n\pi}\right)\hat{g}(n)$$

converges absolutely.

Proof

By Parseval's theorem (technically Bessel's is enough), we've:

$$\sum_{n\in\mathbb{Z}} |\hat{g}(n)|^2 = \|g\|^2 < \infty$$

Therefore, it follows that $\sum|\hat{g}(n)| \in \ell^2(\mathbb{Z})$. Trivially, $a_n = \begin{cases} \frac{1}{2|n|\pi} & n\neq 0 \\ 0 & n=0 \end{cases}$ is also in $\ell^2(\mathbb{Z})$ (p-series).

So, utilizing the fact that $\ell^2(\mathbb{Z})$ is a Hilbert space, we'll use Cauchy-Schwarz:

$$\sum_{n\in\mathbb{Z}} a_n|\hat{g}(n)| \leq \sqrt{\sum_{n\in\mathbb{Z}} a_n^2}\sqrt{|\hat{g}(n)|^2} < \infty$$

Therefore, the series converges absolutely.

□

Theorem 1.2.4: If $f \in C^1(S^1)$, then the Fourier series of f converges absolutely and uniformly to f.

Proof,

Let $f \in C^1(S^1)$. So, by continuous differentiability we've:

$$\hat{f}'(n) = 2n\pi i \hat{f}(n)$$

By the Extra Derivative Lemma, we know that:

$$\sum_{n \neq 0} \frac{\hat{f}'(n)}{2n\pi i}, \tag{$*$}$$

converges absolutely.

Therefore, for $n \in \mathbb{Z} \setminus \{0\}$, we've:

$$\left|\hat{f}(n)e_n(x)\right| \leq \left|\hat{f}(n)\right| = \left|\frac{\hat{f}'(n)}{2n\pi i}\right| = \left|\frac{\hat{f}'(n)}{2n\pi}\right|$$

From $(*)$, we know that the series converge. Thus, from Weierstrass M-test,

$$\sum_{n \in \mathbb{Z}} \hat{f}(n)e_n(x)$$

converges absolutely and uniformly to some function $g \in L^2(S^1)$.

Now, for all $n \in \mathbb{Z}$

$$\widehat{f-g}(n) = 0 \implies f - g = 0$$

a.e.

Thus, the Fourier series of f converges almost everywhere to f.

Finally, by uniform convergence, the Fourier series itself is continuous. So, the almost everywhere convergence can be converted to everywhere convergence.

□

Corollary 1.2.4.1: For $k \geq 1$, if $f \in C^{k(S^1)}$, then for $0 \leq j \leq k-1$, the Fourier series of f^j,

$$f^{(j)}(x) = \sum_{n \in \mathbb{Z}} (2\pi i n)^j \hat{f}(n) e_n(x)$$

converges absolutely and uniformly to f^j. In particular, for $f \in C^\infty(S^1)$, we may take arbitrary term-by-term derivatives of the Fourier series of f.

Proof,

Since, $j \leq k-1 \implies f^{(j)}(x) \in C^1(S^1)$. Thus,

$$f^{(j)}(x) = \sum_{n \in \mathbb{Z}} \hat{f}^{(j)}(n) e_n(x)$$

Since, $f \in C^j(S^1)$, therefore, the Fourier series converges to:

$$\widehat{f^{(j)}}(n) = (2n\pi i)^j \hat{f}(n)$$

Therefore, we get

$$f^{(j)}(n) = \sum_{n \in \mathbb{Z}} (2n\pi i)^j \hat{f}(n) e_n(x)$$

□

1.3 Applications of Fourier Series

1.3.1 Decay of Fourier Coefficients and Smoothness

Theorem 1.3.1.1: For $j \geq 1$, let $f \in L^2(S^1)$ have the property that for some constants K and $p > j + 1$ and for all $n \in \mathbb{Z}$, $n \neq 0$, we have:

$$\hat{f}(n) = O\left(\frac{1}{|n|^p}\right)$$

Then the Fourier series of f converges absolutely and uniformly to some $g \in C^j(S^1)$ such that:

$$g^j(x) = \sum_{n \in \mathbb{Z}} (2\pi i n)^j \hat{f}(n) e_n(x)$$

and $f = g$ almost everywhere.

Proof,

Consider,

$$g_\ell(x) = \sum_{n \in \mathbb{Z}} (2\pi i n)^\ell \hat{f}(n) e_n(x)$$

Where, $0 \leq \ell \leq j$. Now,

$$\left|(2\pi i n)^\ell \hat{f}(n) e_n(x)\right| = (2\pi |n|)^\ell \left|\hat{f}(n)\right| \leq (2\pi |n|)^\ell \frac{K}{|n|^p}$$

Where, $K \in \mathbb{R}^+$ is a constant which follows from the fact that $\hat{f}(n) = O\left(\frac{1}{|n|^p}\right)$.

Since, $p - \ell \geq p - j > j + 1 - j = 1$. By the p-series test, the series with the terms on LHS converges. Thus, by Weierstrass M-test, the series converges absolutely and uniformly.

Since, g_0 has the same Fourier coefficients as f, by the Isomorphism of Fourier series, it follows that:

$$f = g_0$$

almost everywhere.

Let,

$$g = g_0$$

We can use the definition of the derivative to get:

$$\lim_{h\to 0} \frac{1}{h} \sum_{n\in\mathbb{Z}} \Big(\hat{f}(n) e_n(x+h) - \hat{f}(n) e_n(x) \Big) = \lim_{h\to 0} \frac{1}{h} \sum_{n\in\mathbb{Z}} \hat{f}(n)(e_n(x+h) - e_n(x))$$

By MVT:

$$e_n(x+h) - e_n(x) = (2n\pi i) e_n(x+\xi h) h$$

where, $\xi \in (0,1)$.

By uniform continuity, we can interchange sum and the limit. And then using continuity, we can conclude:

$$\begin{aligned}
\lim_{h\to 0} \frac{1}{h} \sum_{n\in\mathbb{Z}} \hat{f}(n)(2n\pi i h) e_n(x+\xi h) &= \sum_{n\in\mathbb{Z}} \lim_{h\to 0} \frac{\hat{f}(n)(2n\pi i h) e_n(x+\xi h)}{h} \\
&= \sum_{n\in\mathbb{Z}} (2n\pi i) \hat{f}(n) \lim_{h\to 0} e_n(x+\xi h) \\
&= \sum_{n\in\mathbb{Z}} (2n\pi i) \hat{f}(n) e_n(x)
\end{aligned}$$

Therefore, $g' = g_1$.

We can argue similarly to show:

$$g^{(\ell)} = g_\ell$$

Since the only hypothesis we needed was uniform convergence and continuity of the series in the LHS.

□

Corollary 1.3.1.1.1: Let $f \in L^2(S^1)$ have the property that for all $p > 2$, there exists some constant K_p such that all $n \in \mathbb{Z}, n \neq 0$, we

have $\left|\hat{f}(n)\right| \leq \frac{K_p}{|n|^p}$. Then the Fourier series of f converges uniformly to some $g \in C^\infty(S^1)$ such that $f = g$ a.e.

Proof,

We know that if, $\hat{f}(n) = O\left(\frac{1}{|n|^p}\right)$, we've that:

$$f = g$$

almost everywhere where, $g \in C^{p-2}(S^1)$. Since, p is arbitrary, it follows that we can figure out another g' such that $g' \in C^n(S^1)$ where $n > p - 2$.

Because of Isomorphism of Fourier series, the fact that g, g' have the same Fourier coefficient they're equal almost everywhere. But by the additional condition of continuity, they're equal everywhere.

So, for any n,

$$f = g \text{ a.e. } g \in C^n(S^1) \iff f = g \text{ a.e. } g \in C^\infty(S^1)$$

□

Corollary 1.3.1.1.2: The space $C^\infty(S^1)$ is dense in $L^2(S^1)$

Proof,

For a space to be dense in $L^2(S^1)$, it must follow that for any $f \in L^2(S^1)$, there exists a $g \in C^\infty(S^1)$ such that:

$$\|f - g\|_{L^2} < \varepsilon$$

where, $\varepsilon \in \mathbb{R}^+$ is arbitrary.

Of course, this immediately follows from the fact that the Fourier polynomials of f converge to f in the L^2 norm.

The key realization here is simply that Fourier polynomials are smooth.

□

Theorem 1.3.1.2 (Weierstrass Approximation Theorem): For every $f \in C^0\left(\left[0, \frac{1}{2}\right]\right)$ and $\varepsilon \in \mathbb{R}^+$, there exists some polynomial $p(x)$ such that:

$$|f(x) - p(x)| < \varepsilon$$

For all $x \in \left[0, \frac{1}{2}\right]$

Proof,

Let $\varepsilon \in \mathbb{R}^+$.

Since, f is not in S^1, we consider the even extension. Why even? Because we can't guranteee the continuity of the odd extension without further conditions on f.

Now, let

$$g(x) = f_{\text{even}}(x) = \begin{cases} f(x) & x \geq 0 \\ f(-x) & x < 0 \end{cases}$$

We since $g \in C^0(S^1)$, it follows that its Cesaro sum of its Fourier series converges to it uniformly i.e.,

$$\lim_{N \to \infty} \sup_{x \in [-\frac{1}{2}, \frac{1}{2}]} |g - F_N * g| = 0$$

Since,

$$F_N * g(x) = \frac{1}{N} \sum_{n=0}^{N-1} g_n(x)$$

Because of the fact that g is an real, even function, it follows that:

$$g_n(x) = \sum_{k=0}^{n} a_k \cos(2k\pi x)$$

So, $F_N * g$ is a cosine series of degree at most $N - 1$.

Recall from Taylor's theorem, we can approximate $\cos(2k\pi x)$ uniformly by real polynomials on $\left[-\frac{1}{2}, \frac{1}{2}\right]$.

The reasonining for that is simply due to the fact:

$$\cos(2k\pi x) = \sum_{i=0}^{K} (-1)^i \frac{(2k\pi x)^{2i}}{(2i)!} + R_K(x)$$

where the remainder R_K satisfies:

$$R_K(x) = \int_0^x \mathrm{d}t \, \frac{(x-t)^K}{K!} g^{(k+1)}(t) = \int_0^x \mathrm{d}t \, \frac{(x-t)^K}{K!} \frac{\mathrm{d}^{K+1}}{\mathrm{d}x^{K+1}}(\cos(2k\pi t))$$

Taking absolute values, and integrating, we get the bound:

$$|R_K(x)| \leq \frac{(2k\pi)^{K+1}}{(K+1)!} |x|^{K+1}$$

Taking the supremum over $\left[-\frac{1}{2}, \frac{1}{2}\right]$, we get:

$$\sup_{x \in \left[-\frac{1}{2}, \frac{1}{2}\right]} |R_K| \leq \frac{(2k\pi)^{K+1}}{(K+1)!}$$

(For simplification we used $|x| < 1$, we could've used the sharper bound $|x| \leq \frac{1}{2}$).

Due to the fact that bound on the remainder vanishes as $K \to \infty$, and is uniform, we can conclude uniform convergence.

Now,

$$F_N * g(x) = \sum_{n=0}^{N-1} a_n \cos(2n\pi x)$$

So,

$$p(x) = \sum_{n=0}^{N-1} a_n \sum_{k=0}^{K_n} (-1)^k \frac{(2n\pi x)^{2k}}{(2k)!}$$

Thus,

$$|F_N * g(x) - p(x)| \leq \sum_{n=0}^{N-1} |a_n| \left| R_{K_n}(x) \right|$$

From the above argument, it follows there exists a K_n such that,

$$\sup_{x \in [-\frac{1}{2}, \frac{1}{2}]} \left| R_{K_n}(x) \right| \leq \frac{\varepsilon}{(|a_n| + 1)(N)}$$

for each n.

Using these $\{K_n\}_n$, we get a uniform bound:

$$\begin{aligned}
\sup_{x \in [-\frac{1}{2}, \frac{1}{2}]} |F_N * g(x) - p(x)| &\leq \sum_{n=0}^{N-1} |a_n| \sup_{x \in [-\frac{1}{2}, \frac{1}{2}]} \left| R_{K_n}(x) \right| \\
&\leq \sum_{n=0}^{N-1} |a_n| \frac{\varepsilon}{(|a_n| + 1)(N)} \\
&< \sum_{n=0}^{N-1} \frac{\varepsilon}{N} = \varepsilon
\end{aligned}$$

Thus, we have uniform convergence.

With this in our toolkit, the remaining argument remains simple.

$$|g(x) - p(x)| \leq |g(x) - F_N * g(x)| + |F_N * g(x) - p(x)|$$

Now, N, p are independent choices. So, simply choose N such that the first term in the RHS is $< \frac{\varepsilon}{2}$ uniformly, then pick p such that the second term in RHS is $< \frac{\varepsilon}{2}$ uniformly. Both follow from uniform convergence.

These bounds are also uniform since:

$$\begin{aligned}
\sup|g - p| &\leq \sup|g - F_N * g| + \sup|F_N * g - p| \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
&= \varepsilon
\end{aligned}$$

Thus, g can be uniformly approximated by polynomials on $\left[-\frac{1}{2}, \frac{1}{2}\right]$. Since, f is merely the restriction of g on $\left[0, \frac{1}{2}\right]$, the uniform bound means f can also be uniformly approximated.

□

Theorem 1.3.1.3 (Existence of the Bump function): For $a < b, \delta > 0$, there exists some $\varphi : \mathbb{R} \to \mathbb{R}$ such that:

1. $\varphi \in C^\infty(\mathbb{R})$
2. $x \in [a, b] \implies \varphi(x) = 1$
3. $x \in [a - \delta, a], x \in [b, b + \delta] \implies \varphi(x) \in [0, 1]$
4. $x \in \mathbb{R} \setminus (a - \delta, b + \delta) \implies \varphi(x) = 0$

Proof,

Lemma 1.3.1.4: Define,

$$\varphi_1(x) = \begin{cases} e^{-\frac{1}{x}} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

We claim $\varphi_1 \in C^\infty(\mathbb{R})$.

This result is trivial on $\mathbb{R} \setminus \{0\}$ because both 0 and $e^{-\frac{1}{x}}$ are infinitely differentiable.

So, we have to analyze $x = 0$.

Now, we first prove by induction that the form of $\varphi_1^{(n)}$ is of the form for $x > 0$

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\varphi_1(x) = \frac{e^{-\frac{1}{x}}}{x^{n+1}} p_{n-1}\left(\frac{1}{x}\right)$$

where, p is a $n - 1$ degree real polynomial.

It should be clear that why this form implies infinite differentiability of φ_1 for $x > 0$.

Now, for $n = 1$, the base case, we've:

$$\varphi_1{}'(x) = \frac{e^{-\frac{1}{x}}}{x^2}(1)$$

This means the base case is valid.

Assume that for $k \geq n$,

$$\frac{\mathrm{d}^k}{\mathrm{d}x^k}\varphi_1(x) = \frac{e^{-\frac{1}{x}}}{x^{k+1}}p_{k-1}\left(\frac{1}{x}\right)$$

for $x > 0$.

Differentiating φ_1 $k+1$ times wrt to x, we get:

$$\begin{aligned}\frac{\mathrm{d}^{k+1}}{\mathrm{d}x^{k+1}}\varphi_1(x) &= \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{e^{-\frac{1}{x}}}{x^{k+1}}p_{k-1}\left(\frac{1}{x}\right)\right] \\ &= \frac{e^{-\frac{1}{x}}}{x^{k+2}}\left[-(k+1)p_{k-1}\left(\frac{1}{x}\right) + \frac{1}{x}p_{k-1}\left(\frac{1}{x}\right) - \frac{1}{x}p'_{k-1}\left(\frac{1}{x}\right)\right]\end{aligned}$$

Since,

$$-(k+1)p_{k-1}\left(\frac{1}{x}\right) + \frac{1}{x}p_{k-1}\left(\frac{1}{x}\right) - \frac{1}{x}p'_{k-1}\left(\frac{1}{x}\right) = p_k\left(\frac{1}{x}\right)$$

where, p_k is a kth degree polynomial, it follows that:

$$\frac{\mathrm{d}^{k+1}}{\mathrm{d}x^{k+1}}\varphi_1(x) = \frac{e^{-\frac{1}{x}}}{x^{k+2}}p_k\left(\frac{1}{x}\right)$$

Thus, we're done.

Remark: We could compute out the polynomial $p_k(x)$ since we know it follows the Recurrence:

$$p_k(x) = xp_{k-1}(x) - xp'_{k-1}(x) - (k-1)p_{k+1}(x)$$

Now, since,

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\varphi_1(x) = \frac{e^{-\frac{1}{x}}}{x^{n+1}}p_{n-1}\left(\frac{1}{x}\right)$$

For $x > 0$.

We also know that, rather trivially,

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\varphi_1(x) = 0$$

for $x < 0$.

So,

$$\lim_{h\to 0^+} \frac{1}{h}\left[\frac{e^{-\frac{1}{h}}}{h^{n+1}}p_{n-1}\left(\frac{1}{h}\right)\right] = \lim_{h\to\infty} h^{n+2}p_{n-1}(h)e^{-h} = 0$$

Because of basic asymptotics (exponential $\gg$ polynomial).

In the other direction, it's trivially zero because the numerator of the difference quotient is $\frac{\mathrm{d}^n}{\mathrm{d}x^n}\varphi_1(x) = 0$.

Thus,

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\varphi_1^n(0) = 0$$

For any $n \in \mathbb{N}$.

Thus, $\varphi_1 \in C^\infty(\mathbb{R})$.

Lemma 1.3.1.5: $\varphi_2 : \mathbb{R} \to \mathbb{R}$ defined by:

$$\varphi_2(x) = \varphi_1(x)\varphi_1(1-x)$$

is a C^∞ function such that:

$$\varphi_2(x) > 0 \quad x \in (0,1) \quad \varphi_2(x) = 0 \quad \text{otherwise}$$

Within $x \in (0,1)$ both terms $x, 1-x > 0$, thus $\varphi_2 > 0$. However, the moment we go outside, one of $x, x-1$ becomes negative. Hence, $\varphi_2(x) = 0\varphi_1(1-x)$ or $\varphi_2(x) = \varphi_1(x)0$. Thus,

$$\varphi_2(x) = 0 \quad x \in \mathbb{R} \setminus (0,1)$$

C^∞ follows from φ_2 being a product of the smooth functions $\varphi_1(x), \varphi_1(1-x)$.

Finally, we have our penultimate lemma:

Lemma 1.3.1.6: Let, $A = \int_0^1 \mathrm{d}x\, \varphi_2(x)$. Then,

$$\varphi_3(x) = \frac{1}{A} \int_0^x \mathrm{d}t\, \varphi_2(t)$$

is an increasing C^∞ function such that $\varphi_3(x)$ for $x \leq 0$ and $\varphi_3(x) = 1$ for $x \geq 1$

φ_3 is a smooth step function.

Recall that:

$$\varphi_2(x) = \begin{cases} > 0 & x \in (0,1) \\ 0 & x \in \mathbb{R} \setminus (0,1) \end{cases}$$

Now, for $x \leq 0$, $\varphi_2(x) = 0$, and since the integrand vanishes, it follows $\varphi_3(x) = 0$.

Now, for $x \geq 1$, we can split the integral as follows:

$$\int_{(0,1)} \mathrm{d}t\, \varphi_2(t) + \int_{x \geq 1} \mathrm{d}t\, \varphi_2(t) = A + 0 = A$$

By definition of A.

Thus,

$$\varphi_3(x) = 1 \quad x \geq 1$$

For monotonicity, we differentiate,

$$\varphi_3{}'(x) = \frac{1}{A}\varphi_2(x)$$

Since, the derivative of φ_3 is smooth, it also follows that φ_3 is smooth by definition.

Consider $x, y \in (0,1)$ such that $x < y$. Then, by MVT we've:

$$\varphi_3(y) - \varphi_3(y) = \frac{1}{A}\varphi_2(\xi)(y-x)$$

Since, $\xi \in (0,1) \implies \varphi_2(\xi) > 0$. As, $A > 0 (\because \varphi_2 > 0, x \in (0,1)), y - x > 0$, it follows that:

$$\varphi_3(y) - \varphi_3(x) > 0$$

Thus, φ_3 is increasing.

Now we can prove the theorem. We can construct the bump function from $\varphi_3(x)$.

First note that $\varphi_3(x)$ does it step in 1 unit. Thus,

$$\varphi_3\left(\frac{x}{\delta}\right)$$

Does its step in δ unit.

For the step to start from $a - \delta$, we translate:

$$\varphi_3\left(\frac{x-(a-\delta)}{\delta}\right)$$

Finally, we need to start stepping down from b and the step down should take δ units.

So, we can subtract,

$$\varphi(x) = \varphi_3\left(\frac{x-(a-\delta)}{\delta}\right) - \varphi_3\left(\frac{x-b}{\delta}\right)$$

Here, φ is the required bump function. □

Definition 1.3.1.1 (Riemann Zeta Function): For real $s > 1$, the Riemann Zeta function $\zeta(s)$ is defined by:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

□

Theorem 1.3.1.7: We have that:

$$\zeta(2) = \frac{\pi^2}{6} \quad \zeta(4) = \frac{\pi^4}{90}$$

Proof,

Consider the Fourier series of $|x|$.

$$f : S^1 \to f(x) = |x| \implies \hat{f}(n) = \int_{-\frac{1}{2}}^{\frac{1}{2}} \mathrm{d}x \, |x| \overline{e_n(x)}$$

Computing the integral, we get:

$$\hat{f}(n) = \begin{cases} \frac{(-1)^n - 1}{2n^2\pi^2} & n \neq 0 \\ \frac{1}{2} & n = 0 \end{cases}$$

Thus, the Fourier series of f is:

$$g(x) = \frac{1}{4} + \sum_{n \neq 0} \frac{(-1)^n - 1}{2n^2\pi^2} e_n(x)$$

Because of the fact that:

$$\left| \frac{(-1)^n - 1}{2n^2\pi^2} e_n(x) \right| = O\left(\frac{1}{|n|^2} \right)$$

It follows that the g converges uniformly on S^1.

Now, by uniform convergence,

$$\begin{aligned}\widehat{f-g}(n) &= \int_0^1 \mathrm{d}x\,(f-g)(x)\overline{e_n(x)} \\ &= \int_0^1 \mathrm{d}x\,f(x)\overline{e_n(x)} - \int_0^1 \mathrm{d}x \sum_{m\in\mathbb{Z}} \hat{f}(m)e_m(x)\overline{e_n(x)} \\ &= \hat{f}(n) - \sum_{m\in\mathbb{Z}} \int_0^1 \mathrm{d}x\,\hat{f}(m)e_m(x)\overline{e_n(x)}\end{aligned}$$

Where, the interchange of the sum and the integral is validated by uniform convergence of g.

Simplifying, we get:

$$\widehat{f-g}(n) = \hat{f}(n) - \hat{f}(n) = 0$$

Therefore, as $f-g$'s Fourier coefficients vanish, then:

$$f - g = 0$$

almost everywhere. Finally, the continuity condition on f, implies this almost everywhere convergence can be extended to everywhere equality.

Hence,

$$f = g$$

Now,

$$f(x) = \frac{1}{4} + \sum_{n\neq 0} \frac{(-1)^n - 1}{2n^2\pi^2} e_n(x)$$

We can then simplify this:

$$f(x) = \frac{1}{4} - \sum_{n\in\mathbb{Z}} \frac{e_{2n+1}(x)}{(2n+1)^2\pi^2}$$

Finally, by the symmetry of odd numbers, we can further refine our estimate to:

$$f(x) = \frac{1}{4} - 2\sum_{n=0}^{\infty} \frac{e_{2n+1}(x)}{(2n+1)^2\pi^2}$$

Hence, using $x = 0$, we get:

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$$

Now, recall that we're summing up:

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + ...$$

So, the sum, we're ignoring here is:

$$\frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{6^2}... = \frac{1}{2^2}\left(1 + \frac{1}{2^2} + \frac{1}{3^2}\right) = \frac{1}{2^2}\zeta(2) = \frac{1}{4}\zeta(2)$$

Thus,

$$\frac{\pi^2}{8} + \frac{1}{4}\zeta(2) = \zeta(2) \implies \zeta(2) = \frac{\pi^2}{6}$$

Finally, using Parseval's:

$$\frac{1}{16} + 4\sum_{n=0}^{\infty} \frac{1}{(2n+1)^4\pi^4} = \|\|x\|\| \implies \frac{\pi^4}{12} = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^4}$$

Therefore, using the same technique as above:

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

□
